Software Requirements Specification (SRS)

Project Title

AI-Based Automatic Waste Classification System

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1. Introduction

1.1 Purpose

The purpose of this project is to develop a computer vision-based system that automatically identifies and classifies waste into three categories: Recyclable, Organic, and Hazardous. The system aims to improve waste segregation, reduce manual effort, and promote environmentally sustainable waste management using Artificial Intelligence and Deep Learning.

1.2 Scope

The system accepts waste images through a webcam or image upload, processes the images, classifies the waste using a trained deep learning model, and displays the predicted category with a confidence score. The system can also maintain a prediction history for future analysis.

1.3 Intended Audience

- Project Developers
- Project Guide
- Software Testers
- End Users

1.4 Definitions

Term	Description
AI	Artificial Intelligence
DL	Deep Learning
YOLOv8	You Only Look Once Version 8 – real-time deep learning model used for waste image classification

Term	Description
FastAPI	Python web framework used to serve the trained YOLOv8 model
Dataset	Collection of labeled waste images used to train the YOLOv8 classification model
Classification	Identifying the waste category as Recyclable, Organic, or Hazardous
Confidence Score	Probability value indicating the model's confidence in its prediction

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2. Overall Description

2.1 Product Overview

The proposed system is an AI-based application that classifies waste images into three categories using a YOLOv8 Classification model deployed through FastAPI.

Project Flow

Capture/Upload Image

↓

Image Preprocessing

↓

YOLOv8 Classification Model

↓

Waste Classification

(Recyclable / Organic / Hazardous)

↓

Display Prediction

↓

Store Prediction History

2.2 Product Functions

The system shall:

- Capture images using a webcam.
- Upload waste images.
- Preprocess images.
- Classify waste into three categories.
- Display prediction results with confidence.
- Store prediction history**.**

2.3 User Classes

Administrator

- Manage datasets
- Train the model
- View prediction reports

User

- Upload or capture waste images
- View classification results

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3. Functional Requirements

FR-1 Image Upload

The system shall accept image formats:

- JPG
- JPEG
- PNG

FR-2 Camera Capture

The system shall capture waste images through a webcam.

FR-3 Image Preprocessing

The system shall preprocess images by:

- Resizing
- Normalization

- Noise removal

FR-4 Waste Classification

The trained YOLOv8 Classification model shall classify waste into:

- Recyclable
- Organic
- Hazardous

FR-5 Result Display

The system shall display:

- Waste Category
- Confidence Score
- Prediction Time

FR-6 History Management

The system shall store prediction details for future reference.

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4. External Interface Requirements

User Interface

The application shall provide:

- Home Page
- Upload Image
- Camera Capture
- Prediction Result
- History Page

Hardware Requirements

- Webcam
- Computer/Laptop
- GPU (Optional)

Software Requirements

- Python

- OpenCV
- PyTorch (Ultralytics YOLOv8)
- NumPy
- Pandas
- FastAPI
- SQLite/MySQL

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5. Non-Functional Requirements

Performance

- Prediction time should be less than 2 seconds.

Accuracy

- Classification accuracy should be above 90%.

Reliability

- The system should provide consistent and reliable predictions.

Security

- Administrator access should be password protected.
- Prediction records should be securely stored.

Portability

- The application should run on Windows and Linux platforms.

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6. System Model

Use Case Diagram

```
              User
               │
   ┌───────────────────────────────┐
   │              │             │
Upload Image   Capture Image   View Result
   │              │
   └────────────────────────────────┘
               │
         AI Classification
               │
   ┌─────────────────────────────┐
   │             │              │
Recyclable     Organic       Hazardous
```

Data Flow Diagram (Level 0)

```
                User
                 │
                 ▼
        Capture/Upload Image
                 │
                 ▼
         Image Preprocessing
                 │
                 ▼
  YOLOv8 Image Classification Model
                 │
```

```
▼
Waste Classification
│
▼
Display Prediction Results
(Waste Category + Confidence Score
+ Disposal Method)
│
▼
User Dashboard
```

7. Data Requirements

Input

- Image Formats: JPG, JPEG, PNG
- Image Size: 224 × 224 pixels

Output

- Waste Category
- Confidence Score
- Prediction Time

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8. Assumptions and Constraints

Assumptions

- Images are clear and properly captured.
- Dataset is correctly labeled.
- The AI model is trained before deployment.

Constraints

- Poor lighting may reduce prediction accuracy.
- Only three waste categories are supported.
- Internet is required only for cloud deployment.

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9. Future Enhancements

- Mobile application
- Smart dustbin integration
- IoT-based waste monitoring
- Multi-object detection using YOLO
- Cloud dashboard
- Voice assistance

Conclusion

The AI-Based Automatic Waste Classification System provides an intelligent solution for automatic waste segregation using Computer Vision and Deep Learning. It accurately classifies waste into Recyclable, Organic, and Hazardous categories, reducing manual effort and improving recycling efficiency. This SRS document defines the essential requirements for the successful design, development, testing, and deployment of the system.